

Elementary two-fingered robotic arm control

Foteinos Konstantinos ({it2023086,kfoteinos}@hua.gr)

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1 Problem statement

The current work aims to accomplish a naive pick-and-place task, implementing a basic inverse kinematics solver and a controller from scratch. As shown in Fig. 1, the arm is asked to grasp the blue box and throw it into the basket. In particular, the motion planning has been performed manually, that is, the end-effector keyposes have been hard-coded (which requires access to the box and basket poses, which are considered to be known). The task is decomposed into the following sub-goals: 1) open the gripper, 2) place the gripper on top of the object with the desirable end-effector orientation, 3) move the end-effector at the object height, 4) close the gripper, 5) move the gripper on top of the basket, 6) open the gripper to leave the object, 7) return to zero configuration. Sub-goal 2 is essential to avoid collision with the object that may change its pose. The controller does not proceed with a new sub-goal until the last one is accomplished. To determine if a goal is accomplished, the robot compares its internal state (i.e., motor positions) with the desired. Regarding the gripper, if the motor position does not change between two consecutive timesteps, this fact is perceived as the box being grasped, therefore the goal is treated as achieved.

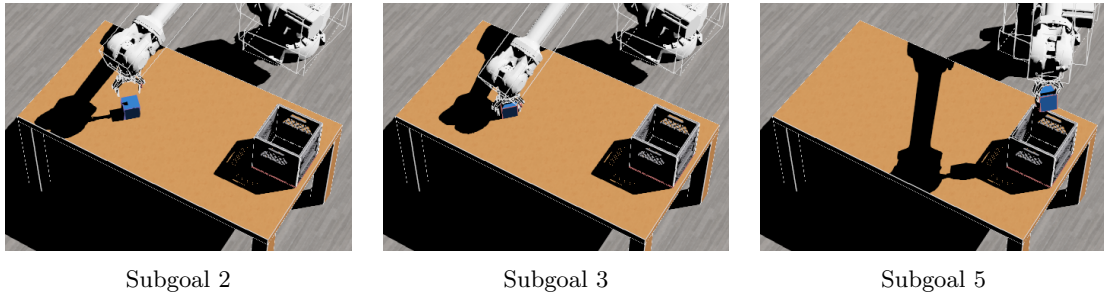


Figure 1: The task considered in the current work.

Repository: https://github.com/foteinosKonstantinos/simple_kinematics

2 Forward kinematics

This section discusses the formulation of forward kinematics and how it was applied in the current work for controlling an IRB 4600/40 robotic arm equipped with a two-finger gripper ROBOTIQ 2F-140. The mathematical formulation is mainly based on [2, 1] with a few modifications.

Conventions: Let $\{0\}$ denote the initial frame, which is considered to be grounded. For any frame $\{a\}$, let $\zeta_{\{a\}}(r) = \zeta_a(r)$ denote the coordinates of the (self-existent, independent of coordinate

choice) vector or point r lying on the 3D space, w.r.t $\{a\}$. Further, let $\hat{x}_a, \hat{y}_a, \hat{z}_a$ denote the unitary vectors (axes) determining the orientation of the frame $\{a\}$. Let o_a denote the origin of $\{a\}$. For instance, $\zeta_a(\hat{y}_a) = (0, 1, 0) \in \mathbb{R}^3$, $\zeta_a(o_a) = (0, 0, 0) \in \mathbb{R}^3$. Note that ζ differs conceptually, depending on whether it is applied to a vector or a point. The vectors analysed here are not tangent, that is, they are not attached to any point.

Rigid motions: The displacement of a frame $\{a\} \rightarrow \{b\}$ is equivalent to the composition of a rotation (preserving orientation) and a translation. These notions can be mathematically described as an orthonormal matrix R_{ab} satisfying $\det R = +1$ and a vector $p_{ab} \in \mathbb{R}^3$ representing the frame axes and origin of the new coordinate system $\{b\}$ w.r.t. to the initial $\{a\}$, respectively. Concretely:

$$R_{ab} = [\zeta_a(\hat{x}_b) \quad \zeta_a(\hat{y}_b) \quad \zeta_a(\hat{z}_b)], p_{ab} = \zeta_a(o_b)$$

Transforming the coordinates of a vector v w.r.t. to $\{b\}$ back to its coordinates w.r.t. to $\{a\}$ can be performed by multiplying by R_{ab} : $\zeta_a(v) = R_{ab} \cdot \zeta_b(v)$. The conceptual discrimination mentioned earlier is shown up here, concerning the transformation of a point r : $\zeta_a(r) = R_{ab} \cdot \zeta_b(r) + p_{ab}$. The matrices R_{ab} constitute a group named special orthogonal group (denoted by $\mathbb{SO}(3)$). For computational convenience, R_{ab} and p_{ab} are normally combined to a homogenous matrix T_{ab} representing the pose of $\{b\}$ w.r.t. $\{a\}$, belonging to the special euclidean group ($\mathbb{SE}(3)$):

$$T_{ab} = \begin{bmatrix} R_{ab} & p_{ab} \\ 0 & 1 \end{bmatrix} \in \mathbb{SE}(3) \subset \mathbb{R}^{4 \times 4}$$

With this formula, the change of coordinates can be expressed as follows:

$$\begin{bmatrix} \zeta_a(r) \\ 1 \end{bmatrix} = T_{ab} \cdot \begin{bmatrix} \zeta_b(r) \\ 1 \end{bmatrix}$$

Obviously, $T_{ab} = T_{ba}^{-1}$. The following equation holds and it will be proved very useful later:

$$T_{a_0 a_n} = \prod_{i=0}^{n-1} T_{a_i a_{i+1}}$$

for any frames $a_i, i \in \{0, 1, \dots, n\}$, as intuition deduces.

Angular velocities: It can be proved that if $\{b\}$ is rotating with (unitary) angular velocity $\hat{\omega}$, then $\frac{d}{dt}(R_{ab}) = \zeta_a(\hat{\omega}) \times R_{ab}$, where the cross product is applied to every column. If instead of $\hat{\omega}$ an antisymmetric matrix $[\hat{\omega}]$ is defined with similar behaviour (which can always be done), then

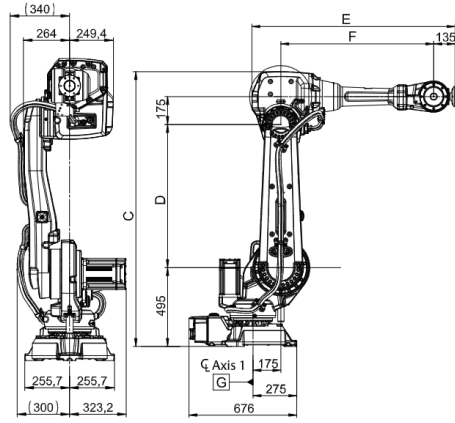
$$\frac{d}{dt}(R_{ab}) = [\zeta_a(\hat{\omega})] \cdot R_{ab} \implies R_{ab}(t) = \exp([\zeta_a(\omega)]t) R_{ab}(0)$$

Given that the speed is one, after θ time units the frame must have been rotated by θ radians around the axis $\hat{\omega}$, starting from the initial frame $\{a\}$ (i.e., $R_{ab}(0) = \mathbb{I}_{3 \times 3}$). Therefore,

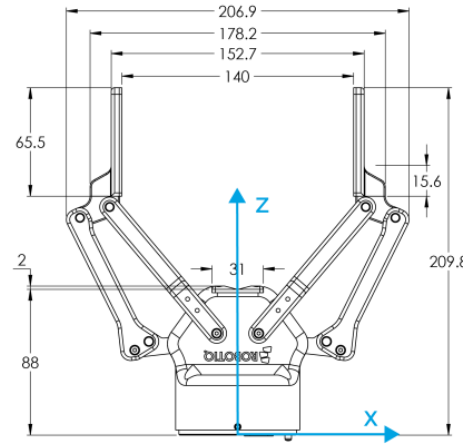
$$R_{ab} = \exp([\zeta_a(\hat{\omega})\theta])$$

More details on Lie algebra can be found in the references [2].

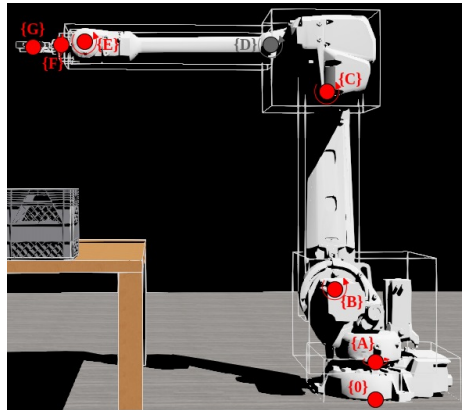
Application: In this study, a hybrid formulation is adapted for simplicity¹, where frames are attached to the revolute joints (and therefore to the child links) of an IRB 4600/40 robotic arm with gripper. Starting from the frame of the joint connecting the robot with the ground (denoted by $\{0\}$), which is also considered to be identical to the frame of the base link, the other frames are defined sequentially (e.g., $\{A\}$ for the A joint, as in the Webots documentation). Then the forward kinematics, given the rotation axes of the joints, their angles (i.e., the motor positions) and their relative positions, can be computed using the above product and exponential formulas. The



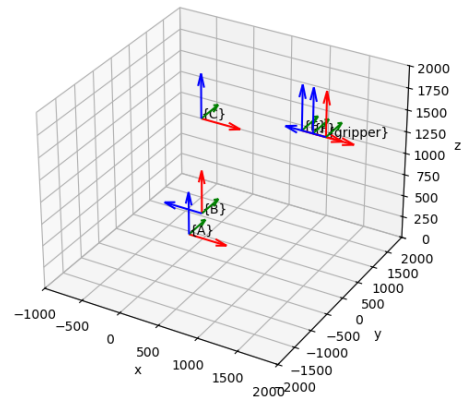
IRB 4600



ROBOTIQ 2F-140



Joints and corresponding frames



Frames in home configuration

Figure 2: Figures from the specifications and the kinematics model.

Frame	Rotation axis $\hat{\omega}$	Angle θ (in radians)	Origin position $\zeta(o)$ (in mm)
T_{0A}	(0, 0, 1)	Motor A	(0, 0, 200)
T_{AB}	(0, 1, 0)	Motor $B -\pi/2$	(175, 0, 295)
T_{BC}	(0, 1, 0)	Motor $C +\pi/2$	(1095, 0, 0)
T_{CE}	(0, 1, 0)	Motor E	(1270, 0, 175)
T_{EF}	(1, 0, 0)	Motor F	(135, 0, 0)
T_{FG}	(0, 1, 0)	$-\pi/2$	(160, 0, 0)

Table 1: The frames considered in the current work, expressed w.r.t. the previous ones. In the most cases, θ depends on the motor position.

following figure (Fig. 2) illustrates the structure of the robot (at zero configuration). The first two pictures are borrowed from the IRB 4600 and ROBOTIQ 2F-140 specifications.

Assumptions and kinematics model: The D joint and the corresponding frame are ignored, as the information regarding their pose is insufficient. The distance of $\{A\}$ from the ground is not provided, therefore it is set approximately equal to $20cm$. The following table (Table 1) reports the frames considered in this implementation. The joints B and C are treated as being on the $\hat{z}_0$ axis in the home configuration, as this simplification does not affect the rest of the analysis. The gripper pose (T_{FG}) is set to be at $160mm$ distance (i.e., near to the grasping point) and with the $\hat{z}_G$ axis at the $-\hat{x}_0$ direction, for convenient guidance.

3 Inverse kinematics

Given a target pose $T_{target} \in \mathbb{SE}(3) \subset \mathbb{R}^{4 \times 4}$, a cost function is minimized w.r.t. to joint angles (denoted by a, b, c, e, f) under constraints, imposed by the underlying hardware. In particular, the cost is defined as follows:

$$L(T_{target}, a, b, c, e, f) = \sum_{i < 4, j} (\mathbb{1}(j < 4) \cdot (\gamma - 1) + 1) \cdot ((T_{0G}(a, b, c, e, f))_{i,j} - (T_{target})_{i,j})^2$$

where γ aims to balance terms coming from orientation error ($j < 4$) and from position error ($j = 4$) (as p_{0G} is expressed in mm we set $\gamma = 1000$), $T_{0G}(a, b, c, e, f)$ is the gripper pose of the considered configuration, and $\mathbb{1}$ is the indicator function. Consequently, inverse kinematics are turned into the following (constrained) optimization problem:

$$a^*, b^*, c^*, e^*, f^* = \arg \min_{a, b, c, e, f \in \mathcal{J}} L(T_{target}, a, b, c, e, f)$$

where $\mathcal{J} \subset \mathbb{R}^5$ contains the allowed motor positions.

References

- [1] Kevin M Lynch and Frank C Park. Modern robotics. *Mechanics, Planning, and Control*, 2017.
- [2] Joan Sola, Jeremie Deray, and Dinesh Atchuthan. A micro lie theory for state estimation in robotics. *arXiv preprint arXiv:1812.01537*, 2018.

¹As mentioned in [1], the screw-theoretic forward kinematics approach is the most dominant.